

① a)

$$DXF = \begin{vmatrix} e_1 & e_2 & e_3 \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x \sin y + e^{3z} & y + x^2 \cos y & 3x e^{3z} - 1 \end{vmatrix}$$

$$= (0 - 0, 3e^{3z} - 3e^{3z}, 2x \cos y - 2x \cos y) = (0, 0, 0) \text{ na } \mathbb{R}^2.$$

$\Rightarrow F$ je potenciálové na $\mathbb{R}^2$.

$$b) \frac{\partial f}{\partial x} = 2x \sin y + e^{3z} \quad (1)$$

$$\frac{\partial f}{\partial y} = y + x^2 \cos y \quad (2)$$

$$\frac{\partial f}{\partial z} = 3x e^{3z} - 1 \quad (3)$$

$$(1) \Rightarrow f(x, y, z) = x^2 \sin y + x e^{3z} + C_1(y, z)$$

$$\text{Dosazením do (2) máme: } x^2 \cancel{\cos y} + \frac{\partial C_1}{\partial y} = y + x^2 \cancel{\cos y}$$

$$\Rightarrow \frac{\partial C_1}{\partial y} = y \Rightarrow C_1(y, z) = \frac{y^2}{2} + C_2(z).$$

$$\text{Odtud } f(x, y, z) = x^2 \sin y + x e^{3z} + \frac{y^2}{2} + C_2(z).$$

$$\text{Dosazením do (3) dostaneme: } 3x e^{3z} + C_2'(z) = 3x e^{3z} - 1$$

$$\Rightarrow C_2'(z) = -1 \Rightarrow C_2(z) = -z + K, K \in \mathbb{R}.$$

$$\text{Položte } 0 = f(1, 0, 0) = 1 + C_2(0) = 1 + K, \text{ je } K = -1.$$

Tedy

$$f(x, y, z) = x^2 \sin y + x e^{3z} + \frac{y^2}{2} - z - 1.$$

$$\textcircled{2} \quad \frac{\partial f}{\partial x} = 3x^2 + y^2 - 12 = 0$$

$$\frac{\partial f}{\partial y} = 2xy + 2y = 0 \dots 2y(x+1) = 0 \begin{cases} y=0 \\ x=-1 \end{cases}$$

$$\bullet y=0 \Rightarrow 3x^2 = 12$$

$$x = \pm 2$$

$$\bullet x = -1 \Rightarrow y^2 = 9$$

$$y = \pm 3$$

Stacionárne body f sú: $A = (2, 0)$, $B = (-2, 0)$, $C = (-1, 3)$,
 $D = (-1, -3)$

Hessova matice:

$$H_f(x, y) = \begin{pmatrix} 6x & 2y \\ 2y & 2x+2 \end{pmatrix}$$

$$\bullet H_f(A) = \begin{pmatrix} 12 & 0 \\ 0 & 6 \end{pmatrix} \text{ je pozit. def. } \Rightarrow A \text{ je bod lok. min.}$$

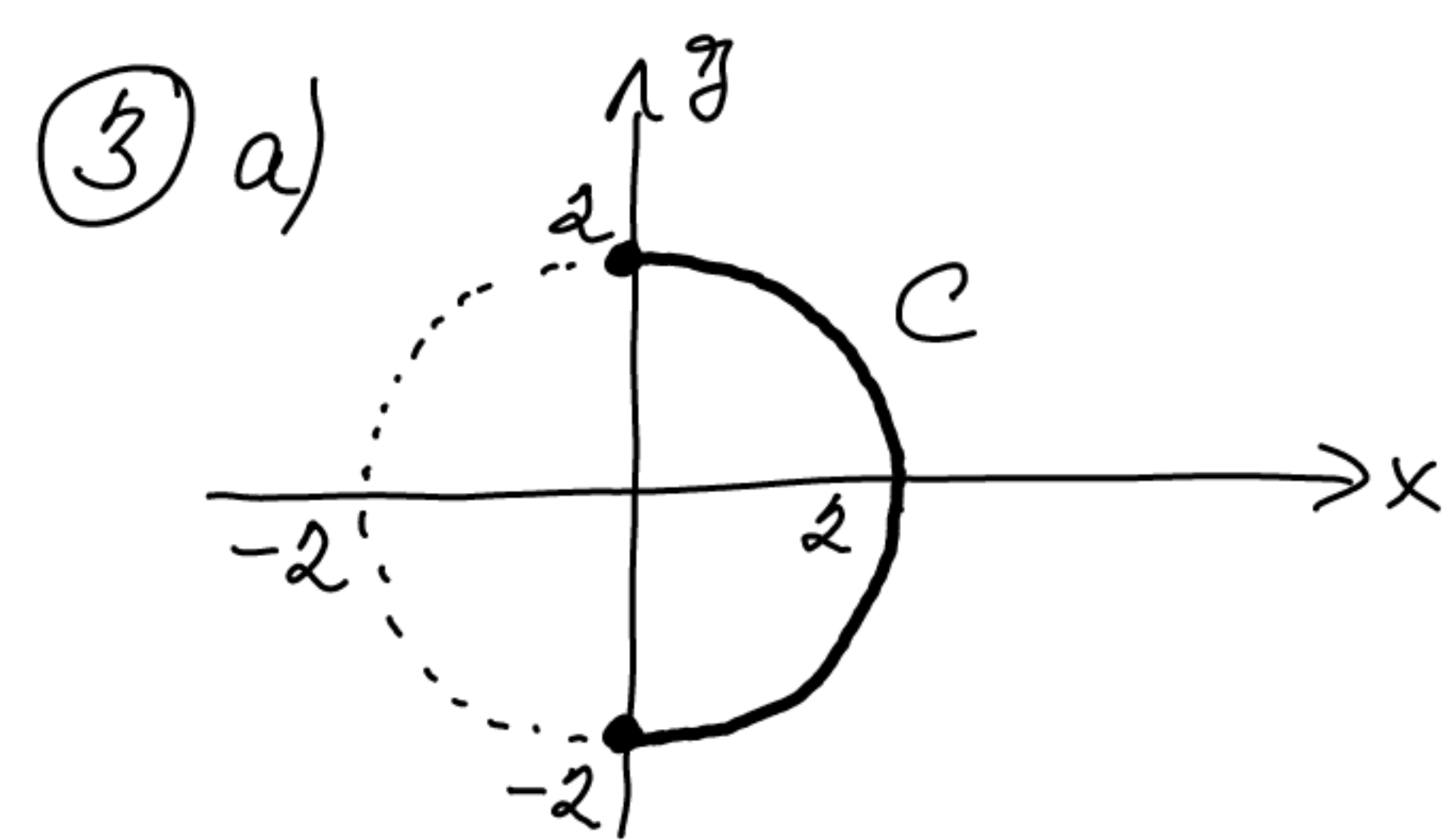
$$\bullet H_f(B) = \begin{pmatrix} -12 & 0 \\ 0 & -2 \end{pmatrix} \text{ je negat. def. } \Rightarrow B \text{ je bod lok. max.}$$

$$\bullet H_f(C) = \begin{pmatrix} -6 & 6 \\ 6 & 0 \end{pmatrix} \dots \det H_f(C) = -36 < 0 \Rightarrow$$

$\Rightarrow H_f(C)$ je indefinitná $\Rightarrow C$ je sedlový bod

$$\bullet H_f(D) = \begin{pmatrix} -6 & -6 \\ -6 & 0 \end{pmatrix} \dots \det H_f(D) = -36 < 0 \Rightarrow$$

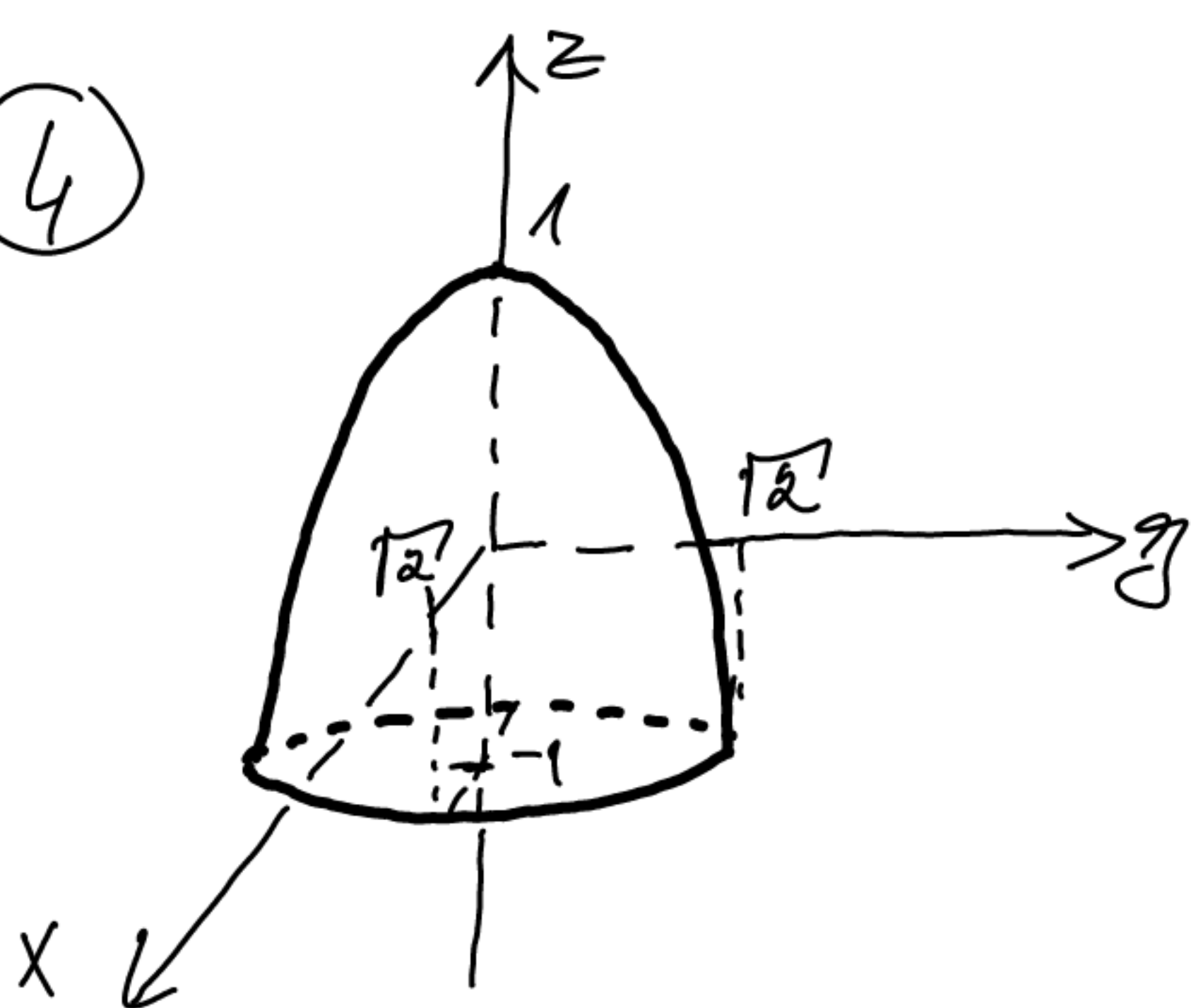
$\Rightarrow H_f(D)$ je indefinitná $\Rightarrow D$ je sedlový bod.



$$C \dots \varphi(t) = (2 \cos t, 2 \sin t) \\ t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right].$$

$$\begin{aligned} b) \int_C x^2 + y &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (4 \cos^2 t + 2 \sin t) \sqrt{4 \cos^2 t + 4 \sin^2 t} dt \\ &= 8 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 t dt + 4 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin t dt \\ &= 16 \int_0^{\frac{\pi}{2}} \frac{1 + \cos(2t)}{2} dt = 8 \frac{\pi}{2} = 4\pi \end{aligned}$$

④



$$V = \{(x, y, z) \mid -1 \leq z \leq 1 - x^2 - y^2\}$$

$$\nabla \cdot F = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = 1 + 2y + 2z$$

$$\int_S F = \int_V \nabla \cdot F = \int_V 1 + 2y + 2z$$

$$= \int_{-1}^1 \int_0^{2\pi} \int_0^{\sqrt{1-z^2}} (1 + 2r \sin \varphi + 2z) r \, dr \, d\varphi \, dz$$

$$= 2\pi \int_{-1}^1 \int_0^{\sqrt{1-z^2}} r(1 + 2z) \, dr \, dz = 2\pi \int_{-1}^1 \frac{1-z^2}{2} (1 + 2z) \, dz$$

$$= \pi \int_{-1}^1 1 + z - 2z^2 \, dz = \pi \left(2 + 0 - \frac{2}{3} \cdot 2 \right)$$

$$= \frac{2\pi}{3}$$

$$\textcircled{5} \text{ a) } \lim_{k \rightarrow \infty} \sqrt[k]{\left| \frac{(x-1)^k}{3^k k} \right|} = \lim_{k \rightarrow \infty} \frac{|x-1|}{3 \sqrt[k]{k}} = \frac{|x-1|}{3} < 1$$

$\Leftrightarrow |x-1| < 3 \dots$ polom. konv. je $R=3$.

Řada diverguje v bodě -3 , neboť
 $|-3-1|=4 > 3$.

$$\text{b) } \sum_{k=1}^{\infty} \frac{(x-1)^k}{3^k k} \stackrel{\text{Der.}}{\sim} \sum_{k=1}^{\infty} \frac{(x-1)^{k-1}}{3^k} = \frac{1}{3} \sum_{k=0}^{\infty} \frac{(x-1)^k}{3^k}$$

$$= \frac{1}{3} \frac{1}{1 - \frac{x-1}{3}} = \frac{1}{4-x}, \quad x \in (-2, 4)$$

$$\Rightarrow \sum_{k=1}^{\infty} \frac{(x-1)^k}{3^k k} = \int \frac{1}{4-x} dx = -\ln(4-x) + C.$$

$$x=1 \dots 0 = -\ln(3) + C \Leftrightarrow C = \ln 3.$$

$$\text{tedy } \sum_{k=1}^{\infty} \frac{(x-1)^k}{3^k k} = -\ln(4-x) + \ln 3, \quad x \in (-2, 4).$$